

OCR Further Mathematics A (2018)

8.01 Recurrence Relations

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1 Preface

1.1 Introduction

A *recurrence system* is a way of describing a sequence by stating how each term is obtained from earlier terms, together with enough information to determine the sequence completely.

Definition 1 (Recurrence relation). *A recurrence relation is an equation that defines each term of a sequence in terms of one or more preceding terms. For example, $u_{n+1} = 3u_n + 2$ and $u_{n+2} = u_{n+1} + u_n$ are recurrence relations.*

Definition 2 (Recurrence system). *A recurrence system consists of a recurrence relation, an initial condition (or conditions), and the range of the variable n . For example,*

$$u_{n+1} = 3u_n + 2, \quad u_0 = 1, \quad n \geq 0.$$

1.2 Types of recurrence relation

A *first-order* linear recurrence relation with constant coefficients has the form

$$u_{n+1} = au_n + f(n),$$

A *second-order* linear recurrence relation with constant coefficients has the form

$$u_{n+2} = au_{n+1} + bu_n + f(n),$$

where a and b are constants.

Definition 3 (Homogeneity). *A recurrence relation is homogeneous if $f(n) = 0$. Otherwise, it is non-homogeneous.*

1.3 Definitions

1.3.1 Term-to-term

Definition 4 (Term-to-term). *A recurrence relation such as $u_{n+1} = au_n + f(n)$ is a term-to-term rule, since it produces the next term from the current one.*

1.3.2 Closed form (position-to-term)

Definition 5 (Closed form (position-to-term)). *A closed form (also called a position-to-term formula) gives u_n directly as a function of n , without reference to any previous term. For example, $u_n = 2^n - 1$ is a closed form. Solving a recurrence system means finding its closed form.*

1.3.3 Complementary function and particular solution

Definition 6 (Complementary function). *The complementary function (CF) is the general solution of the associated homogeneous relation (the relation with $f(n)$ replaced by 0).*

Definition 7 (Particular solution). *A particular solution (PS) is any single sequence satisfying the full relation.*

Definition 8 (General solution). *The general solution is the sum of the complementary function and particular solution.*

2 Solving recurrence relations

To *solve* a recurrence system is to find a formula for u_n in terms of n alone, in the form of a general solution. The complementary function always begins with arbitrary constants, which are then fixed using the initial condition(s).

2.1 First-order recurrence relations

Consider $u_{n+1} = au_n + f(n)$.

Complementary function. Substituting $u_n = x^n$ into the homogeneous relation $u_{n+1} = au_n$ and dividing by x^n gives the auxiliary equation $x - a = 0$, so $x = a$ and

$$\text{CF : } u_n = Aa^n.$$

Particular solution. Try a trial function of the same form as $f(n)$:

- if $f(n)$ is a constant or a polynomial in n , try a polynomial of the same degree;
- if $f(n) = dk^n$, try $u_n = \lambda k^n$.

If the trial function already appears in the complementary function (for instance a constant when $a = 1$, or λk^n when $k = a$), multiply the trial function by n .

Worked example

Solve the recurrence system $u_{n+1} = 3u_n + 4$, $u_0 = 1$, $n \geq 0$, giving u_n in closed form.

Proof. The auxiliary equation is $m - 3 = 0$, giving $m = 3$, so the complementary function is $u_n = A \cdot 3^n$.

Since $f(n) = 4$ is constant and $a = 3 \neq 1$, we try the particular solution $u_n = \lambda$. Substituting,

$$\lambda = 3\lambda + 4 \implies \lambda = -2.$$

The general solution is therefore

$$u_n = A \cdot 3^n - 2.$$

Applying $u_0 = 1$ gives $A - 2 = 1$, so $A = 3$. The closed form is

$$u_n = 3^{n+1} - 2.$$

□

2.2 Second-order recurrence relations

Consider $u_{n+2} = au_{n+1} + bu_n + f(n)$.

Complementary function. Substituting $u_n = x^n$ into the homogeneous relation and dividing by x^n gives the auxiliary equation

$$x^2 - ax - b = 0.$$

The form of the complementary function depends on the nature of its roots:

1. **Distinct real roots** α, β : $u_n = A\alpha^n + B\beta^n$.
2. **Repeated root** α : $u_n = (A + Bn)\alpha^n$.
3. **Complex roots** $p \pm qi$: write them as $r(\cos \theta \pm i \sin \theta)$, where $r = \sqrt{p^2 + q^2}$ and $\tan \theta = q/p$ (with θ taken in the correct quadrant); then

$$u_n = r^n (A \cos n\theta + B \sin n\theta).$$

Particular solution. As in the first-order case, try a function of the same form as $f(n)$, multiplying by n (or n^2 for a repeated root) if it clashes with the complementary function. The general solution is the sum of the complementary function and a particular solution; the two arbitrary constants are then found from the two initial conditions.

Worked example 1

Solve $u_{n+2} = u_{n+1} + 6u_n$, with $u_0 = 1$ and $u_1 = 8$.

Proof. The auxiliary equation is $m^2 - m - 6 = 0$, that is $(m - 3)(m + 2) = 0$, with distinct real roots $m = 3$ and $m = -2$. The relation is homogeneous, so the general solution is

$$u_n = A \cdot 3^n + B(-2)^n.$$

Applying the initial conditions,

$$u_0 = 1 : \quad A + B = 1, \quad u_1 = 8 : \quad 3A - 2B = 8.$$

Solving gives $A = 2$ and $B = -1$, so

$$u_n = 2 \cdot 3^n - (-2)^n.$$

□

Worked example 2

Solve $u_{n+2} = 2u_{n+1} - 2u_n$, with $u_0 = 2$ and $u_1 = 2$.

Proof. The auxiliary equation is $m^2 - 2m + 2 = 0$, with complex roots

$$m = \frac{2 \pm \sqrt{4 - 8}}{2} = 1 \pm i.$$

In modulus-argument form, $r = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\theta = \arctan(1/1) = \frac{\pi}{4}$, so the general solution is

$$u_n = (\sqrt{2})^n \left(A \cos \frac{n\pi}{4} + B \sin \frac{n\pi}{4} \right).$$

Applying $u_0 = 2$ gives $A = 2$. Applying $u_1 = 2$ gives

$$\sqrt{2} \left(A \cos \frac{\pi}{4} + B \sin \frac{\pi}{4} \right) = A + B = 2,$$

so $B = 0$. Hence

$$u_n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}.$$

□

2.3 Trial solutions (ansatz)

$f(n)$	Ansatz
Constant, c	λ
Linear, $an + b$	$\lambda n + \mu$
Quadratic, $an^2 + bn + c$	$\lambda n^2 + \mu n + \nu$
Exponential, dk^n	λk^n

3 Problems

1. Solve the recurrence system $u_{n+1} = 2u_n + 3$, $u_0 = 4$, $n \geq 0$, giving u_n in closed form.
2. Find the closed form of the sequence given by $u_{n+1} = u_n + 5$, $u_1 = 2$, $n \geq 1$.
3. Solve $u_{n+2} = 5u_{n+1} - 6u_n$ with $u_0 = 0$ and $u_1 = 1$.
4. Solve $u_{n+2} = 4u_{n+1} - 4u_n$ with $u_0 = 1$ and $u_1 = 4$. (The auxiliary equation has a repeated root.)
5. Solve $u_{n+2} = 2u_{n+1} - 2u_n$ with $u_0 = 1$ and $u_1 = 1$.
6. Solve the non-homogeneous system $u_{n+2} = 3u_{n+1} - 2u_n + 4$ with $u_0 = 0$ and $u_1 = 0$. (Note that the trial constant clashes with the complementary function.)